DXP-Dependent De Novo Pyridoxal 5′-Phosphate Biosynthesis: A Review-Style Synthesis

1. Executive Summary

Pyridoxal 5′-phosphate (PLP), the catalytically active form of vitamin B6, is an essential cofactor for well over a hundred enzymes spanning amino-acid metabolism, one-carbon chemistry, and neurotransmitter synthesis. Organisms that make PLP *de novo* do so through one of two chemically distinct, non-homologous, and mutually exclusive routes. This review concerns the **deoxyxylulose 5-phosphate (DXP)-dependent** pathway — the longer, six-enzyme route worked out chiefly in *Escherichia coli* and restricted to a subset of eubacteria, most prominently the γ-proteobacteria. It stands in contrast to the shorter, single-complex **DXP-independent PdxS/PdxT (Pdx1/Pdx2)** pathway that predominates across archaea, fungi, plants, protists, and most other bacteria (PMID: 17468224, PMID: 21767669).

Mechanistically, the DXP-dependent route splits into two converging arms. An **erythrose-4-phosphate (E4P) arm** oxidizes and transaminates a four-carbon sugar phosphate through four enzymes — **Epd → PdxB → SerC → PdxA** — to build the amino-ketone **3-amino-2-oxopropyl phosphate**. This intermediate is then condensed with the shared branch-point metabolite **DXP** by the octameric TIM-barrel enzyme **PdxJ** (pyridoxine 5′-phosphate synthase) in an intricate intramolecular ring-closure that assembles the pyridine ring of **pyridoxine 5′-phosphate (PNP)**. Finally, the FMN-dependent oxidase **PdxH (PNPOx)** performs the terminal, O2-dependent oxidation of PNP to PLP (PMID: 11286891, PMID: 15858270).

Two organizing insights emerge from this investigation. First, the pathway is largely **assembled from enzymes borrowed from central metabolism**: Epd is a paralog of glyceraldehyde-3-phosphate dehydrogenase, PdxB belongs to the D-2-hydroxyacid dehydrogenase superfamily, SerC is the serine-biosynthetic phosphoserine aminotransferase, and DXP is manufactured by Dxs for isoprenoid and thiamine metabolism. Only **PdxJ and PdxH are pathway-specialized** in chemistry, and even PdxH is shared with the salvage pathway. Second, the route is **self-referential**: SerC is itself a PLP-dependent enzyme, so the pathway requires its own product to run — a chicken-and-egg dependency that constrains how the pathway can be established evolutionarily. These features define the module's boundaries, its diagnostic genes (the committed early enzymes *epd*, *pdxB*, *pdxA*, *pdxJ*), and its status as a derived, lineage-restricted alternative to the more widely distributed PdxS/PdxT route.

---

2. Definition and Biological Boundaries

What is included

The DXP-dependent de novo PLP biosynthetic system comprises the enzymatic transformations that convert D-erythrose 4-phosphate and DXP into pyridoxal 5′-phosphate. In canonical E. coli order:

Step	Substrate → Product	Enzyme (gene)	EC / activity
1	D-erythrose 4-phosphate → 4-phospho-D-erythronate	**Epd** (*epd*/ *gapB*)	erythrose-4-phosphate dehydrogenase
2	4-phospho-D-erythronate → 3-hydroxy-2-oxo-4-phosphooxybutanoate	**PdxB** (*pdxB*)	4-phosphoerythronate dehydrogenase
3	3-hydroxy-2-oxo-4-phosphooxybutanoate → 4-phosphohydroxy-L-threonine	**SerC** (*serC*/ *pdxF*)	phosphohydroxythreonine aminotransferase
4	4-phosphohydroxy-L-threonine → 3-amino-2-oxopropyl phosphate	**PdxA** (*pdxA*)	4-hydroxythreonine-4-phosphate dehydrogenase (EC 1.1.1.262)
5	3-amino-2-oxopropyl phosphate + DXP → pyridoxine 5′-phosphate	**PdxJ** (*pdxJ*)	pyridoxine 5′-phosphate synthase
6	pyridoxine 5′-phosphate → pyridoxal 5′-phosphate	**PdxH** (*pdxH*)	pyridoxine/pyridoxamine 5′-phosphate oxidase (PNPOx)

Steps 1–4 constitute the **E4P arm** and produce the aminoketone acceptor; step 5 is the **convergent ring-forming condensation** with DXP; step 6 is the **terminal oxidation**.

What lies outside the module — and is commonly confused with it

- **DXP supply (Dxs).** DXP is synthesized by 1-deoxy-D-xylulose-5-phosphate synthase (Dxs), a transketolase-like enzyme condensing (hydroxyethyl)thiamin (from pyruvate) with glyceraldehyde-3-phosphate. DXP is a **shared branch-point metabolite** feeding isoprenoid (MEP pathway), thiamine, and pyridoxol biosynthesis, so its production is placed outside the module boundary (PMID: 9482846).
- **The DXP-independent PdxS/PdxT (Pdx1/Pdx2) pathway.** This is a chemically distinct, non-homologous route in which a single glutamine amidotransferase complex synthesizes PLP directly from ribose 5-phosphate, glyceraldehyde 3-phosphate, and glutamine. It is a **separate module**, not a variant of the DXP-dependent route (PMID: 17159152, PMID: 17468224).
- **Vitamin-B6 salvage.** The recycling of dietary/environmental B6 vitamers (pyridoxal, pyridoxine, pyridoxamine, and their phosphates) via kinases and oxidases is a separate module. Critically, **PdxH is shared between de novo synthesis and salvage**, so its presence in a genome is *not* diagnostic of the DXP-dependent route (PMID: 26823273, PMID: 32253339).
- **Serine biosynthesis.** SerC (phosphoserine aminotransferase) has its canonical role in serine biosynthesis; its recruitment into PLP biosynthesis is a moonlighting/dual-function overlap, not evidence that the two pathways are one system.

Competing definitions

The literature is largely consistent that "DXP-dependent" and "DXP-independent" (also called the "R5P pathway") are the two de novo routes (PMID: 21767669). Ambiguity arises mainly at the **entry step**: whether Epd alone, or Epd together with the housekeeping GapA, provides E4P-oxidizing activity in vivo (see §6). A second definitional subtlety is whether SerC and DXP synthesis "belong" to the pathway; here they are treated as **shared/recruited components** feeding the module rather than as intrinsic, dedicated parts.

---

3. Mechanistic Overview

The best current model is a strictly ordered, convergent pathway in which a four-carbon arm and a five-carbon donor meet at PdxJ:

```
               E4P ARM (steps 1–4)                          DXP ARM (external)
D-erythrose 4-phosphate                                pyruvate + glyceraldehyde-3-P
         | Epd (NAD+)  [1]                                         | Dxs
         v                                                         v
4-phospho-D-erythronate                                1-deoxy-D-xylulose 5-phosphate (DXP)
         | PdxB (NAD+)  [2]                                         |
         v                                                         |
3-hydroxy-2-oxo-4-phosphooxybutanoate                              |
         | SerC (PLP; Glu amino donor) [3]                         |
         v                                                         |
4-phosphohydroxy-L-threonine                                       |
         | PdxA (NAD+, Zn2+) [4]                                   |
         v                                                         |
3-amino-2-oxopropyl phosphate  ----------->  [ PdxJ ] <-------
     (amino-ketone acceptor)        [5] intramolecular condensation / ring closure
                                                 |
                                                 v
                                   pyridoxine 5′-phosphate (PNP)
                                                 | PdxH (FMN, O2) [6]
                                                 v
                                   pyridoxal 5′-phosphate (PLP)
```

Obligatory steps. All six reactions are obligatory for flux from E4P to PLP; the known relationships establish a linear precedence (step 1 → 2 → 3 → 4 → 5 → 6). The order is not merely conventional but **enforced by substrate specificity**: PdxA strictly requires the phosphorylated substrate 4-hydroxy-L-threonine-4-phosphate (PMID: 12896974), and PdxJ only undergoes its catalytic loop closure when *both* DXP and the aminoketone are bound (PMID: 12269807).

Conditional / redundant steps. The entry oxidation (step 1) is partially redundant: both Epd and, to a lesser extent, GapA can oxidize E4P, and only the *gapA epd* double mutant is auxotrophic for pyridoxine (PMID: 9696782). Salvage can bypass the entire de novo route whenever a B6 vitamer is available, rescuing loss-of-function in any de novo gene (PMID: 27060119).

Accessory / shared inputs. DXP (via Dxs) and the amino-group donor for SerC (glutamate) are supplied by central metabolism and are not dedicated to PLP synthesis.

4. Key Findings

Each confirmed finding from the investigation is expanded below with its statistical/biochemical evidence and mechanistic significance.

Finding 1 — Two mutually exclusive de novo PLP pathways; the DXP-dependent route is lineage-restricted

Comparative genomics establishes **two non-homologous de novo PLP biosynthetic routes**: the DXP-dependent six-enzyme pathway (Epd–PdxB–SerC–PdxA–PdxJ–PdxH) characterized in E. coli, and the DXP-independent single-complex PdxS/PdxT (Pdx1/Pdx2) route. The DXP-dependent pathway is restricted to a subset of eubacteria (notably γ-proteobacteria), whereas the DXP-independent pathway predominates across archaea, fungi, plants, protists, and most eubacteria. Animals have **neither** and rely on salvage. As stated directly, "*Two distinct and mutually exclusive de novo pathways have been identified to date, namely deoxyxylulose 5-phosphate dependent, which is restricted to a subset of eubacteria, and deoxyxylulose 5-phosphate independent, present in archaea, fungi, plants, protista, and most eubacteria*" (PMID: 17468224), and "*Pyridoxal phosphate is biosynthesized de novo by two different pathways (the DXP dependent pathway and the R5P pathway) and can also be salvaged from the environment*" (PMID: 21767669). This defines both the module's identity and its boundary against the PdxS/PdxT route and salvage.

Finding 2 — Epd (gapB) is the physiological erythrose-4-phosphate dehydrogenase initiating the pathway

The *gapB*-encoded enzyme of E. coli K-12 was purified and shown to be a **bona fide E4P dehydrogenase (E4PDH), not a second GA3PDH**: "*We found that the gapB-encoded enzyme is indeed an E4PDH and not a second GA3PDH, whereas gapA-encoded GA3PDH used E4P poorly*" (PMID: 7751290). Kinetics: Km(E4P) ≈ 0.96 mM, kcat ≈ 200 s^{-1}, using NAD^+. Genetically, *epd*(*gapB*) mutants contain less PLP and PMP, and *gapA epd* double mutants are pyridoxine auxotrophs, so "*These results implicate the GapA and Epd dehydrogenases in de novo PLP and PMP coenzyme biosynthesis*" (PMID: 9696782). Epd is thus a **recruited paralog** of glycolytic GA3PDH, and the entry step carries partial redundancy.

Finding 3 — PdxJ is an octameric TIM-barrel enzyme that closes the pyridine ring

Crystal structures of E. coli PdxJ (2.0 Å apo and product complexes; 1.96 Å DXP complex) show a **homooctamer** (tetramer of symmetric dimers, 422 symmetry) of TIM-barrel monomers with shared active sites; Arg20 from the partner monomer binds substrate. PdxJ catalyzes "*a complex intramolecular condensation reaction between 1-deoxy-D-xylulose-5′-phosphate and 1-amino-acetone-3-phosphate*" to yield PNP + Pi (PMID: 11286891). The structures reveal multistate substrate binding — "*The octameric enzyme possesses eight distinct binding sites, and three different binding states are observed*" (PMID: 12269807) — and key mechanistic features: "*two phosphate-binding sites with distinct affinities and the existence of a water relay system for the release of reaction water molecules*" (PMID: 12206776). A Schiff-base intermediate and an open–closed catalytic-loop transition (triggered only when both substrates bind) complete the picture. PdxJ is the **committed, pathway-specific ring-forming enzyme**.

Finding 4 — DXP is a shared branch-point metabolite made by Dxs outside the module

The E. coli *dxs* gene encodes a transketolase-like enzyme condensing (hydroxyethyl)thiamin with glyceraldehyde-3-phosphate to form DXP, and "*In E. coli, D-1-deoxyxylulose 5-phosphate is also a precursor for the biosynthesis of thiamin and pyridoxol*" (PMID: 9482846). DXP is therefore the common precursor for isoprenoid (MEP), thiamine, and pyridoxol biosynthesis, justifying its placement outside the module. The PdxA reaction that produces PdxJ's aminoketone acceptor is likewise defined: "*The fourth step is catalyzed by 4-hydroxythreonine-4-phosphate dehydrogenase (PdxA, E.C. 1.1.1.262), which converts 4-hydroxy-l-threonine phosphate (HTP) to 3-amino-2-oxopropyl phosphate*" (PMID: 12896974).

Finding 5 — SerC is a shared, PLP-dependent aminotransferase, making the pathway self-referential

SerC/PSAT (EC 2.6.1.52) is a homodimeric **fold-type I (subgroup IV) PLP-dependent aminotransferase**: "*Phosphoserine aminotransferase (PSAT; EC 2.6.1.52), a member of subgroup IV of the aminotransferases, catalyses the conversion of 3-phosphohydroxypyruvate to l-phosphoserine*" (PMID: 10024454). The E. coli crystal structure (2.3 Å) shows the cofactor bound as an internal aldimine: "*The cofactor is bound through an aldimine linkage to Lys198 in the* *active site*" (PMID: 10024454). Its canonical role is in serine biosynthesis, but the same enzyme performs step 3 of PLP biosynthesis (hence its alternate name PdxF). Because SerC uses PLP as its own cofactor, the pathway is **self-referential** — it needs its product to run.

Finding 6 — PdxB defines the gammaproteobacterial route; the E4P-arm enzymes are recruited paralogs

PdxB (erythronate-4-phosphate dehydrogenase, step 2) belongs to the **D-isomer-specific 2-hydroxyacid dehydrogenase superfamily**; the first crystal structure (*P. aeruginosa*, NAD-bound) shows a homodimer with lid, nucleotide-binding, and C-terminal dimerization domains and half-of-sites-like asymmetric occupancy. As stated, "*One of them is the PdxA/PdxJ pathway found in the gamma subdivision of proteobacteria. It depends on the pdxB gene, which encodes erythronate-4-phosphate dehydrogenase (PdxB), a member of the d-isomer specific 2-hydroxyacid dehydrogenase superfamily*" (PMID: 17217963). The same source notes "*while animals lack any of the pathways for de novo synthesis and salvage of vitamin B6*" (the boundary against animals). Genetic loss of *pdxB* causes B6 auxotrophy rescued by salvage: "*we identified the pdxB gene, encoding erythronate-4-phosphate dehydrogenase, as required for de novo vitamin B6 biosynthesis*" (PMID: 27060119). PdxB is a **diagnostic, committed gene** of the route.

Finding 7 — PdxH/PNPO is a dimeric FMN oxidase performing the terminal O2-dependent oxidation

E. coli PNPOx (PdxH) is a homodimer with a six-stranded antiparallel β-barrel fold and FMN bound at the dimer interface; the *M. tuberculosis* Rv1155 structure (1.8 Å) confirms the fold and FMN-binding residues. It catalyzes the terminal step: "*Escherichia coli pyridoxine 5′-phosphate oxidase (ePNPOx) catalyzes the terminal step in the biosynthesis of pyridoxal 5′-phosphate (PLP) by the FMN oxidation of pyridoxine 5′-phosphate (PNP) or pyridoxamine 5′-phosphate (PMP), forming FMNH(2) and H(2)O(2)*" (PMID: 15858270). The reaction is conserved to humans: "*Rv1155 is a pyridoxine 5′-phosphate oxidase, the Escherichia coli and human counterparts of which catalyse the terminal step in the biosynthesis of pyridoxal 5′-phosphate (PLP)*" (PMID: 16239726). O2 is the terminal electron acceptor, and structures capture an open–closed active-site/tunnel transition.

Finding 8 — PdxH is shared with salvage and is not diagnostic of the DXP-dependent route

In *M. tuberculosis*, de novo PLP biosynthesis proceeds via the DXP-independent (PdxS/PdxT) pathway, yet PdxH enzymes function in salvage: "*De novo biosynthesis of PLP in Mtb takes place through the 'deoxyxylulose 5′-phosphate (DXP)-independent' pathway, whereas PdxH enzymes, possessing pyridoxine/pyridoxamine 5′-phosphate oxidase (PNPOx) activity, are involved in the PLP salvage pathway*" (PMID: 26823273). Therefore the presence of *pdxH* in a genome does **not** indicate operation of the DXP-dependent de novo route; the diagnostic genes are the early/committed ones (*epd*, *pdxB*, *pdxA*, *pdxJ*).

---

5. Major Molecular Players and Active Assemblies

Enzyme	Origin / family	Oligomeric state & cofactor	Status in pathway
Epd	GA3PDH paralog (NAD-dehydrogenase)	NAD^+; Km(E4P)≈0.96 mM, kcat≈200 s^{-1}	Recruited; redundant with GapA
PdxB	D-2-hydroxyacid dehydrogenase superfamily	Homodimer; NAD^+; half-of-sites asymmetry	Committed / diagnostic
SerC	Fold-type I PLP aminotransferase (serine biosynthesis)	Homodimer; PLP (internal aldimine, Lys198)	Shared / dual-function; self-referential
PdxA	Zn-dependent dehydrogenase (EC 1.1.1.262)	Dimer-interface active site; Zn^{2+} (3 His), NAD^+	Committed; strict substrate specificity
PdxJ	Octameric TIM-barrel	Homooctamer (422); Schiff base, water relay	**Pathway-specific** ring-forming enzyme
PdxH	Dimeric FMN oxidase	Homodimer; FMN, O2 acceptor	Shared with salvage; terminal step
(DXP via Dxs)	Transketolase-like	ThDP-dependent	External shared precursor

The take-home is the split between **recruited/shared** components (Epd, PdxB, SerC, PdxH, DXP) and truly **pathway-specific** chemistry (PdxJ). The pathway is essentially built by borrowing dehydrogenase and aminotransferase scaffolds from central metabolism and adding one dedicated ring-forming step (PdxJ) plus a terminal oxidase (PdxH) that is itself co-opted from salvage.

---

6. Evolutionary and Cell-Biological Variation

Distribution across lineages

The two de novo routes are **mutually exclusive and non-homologous**. The DXP-dependent pathway is **lineage-restricted** to a subset of eubacteria, most prominently the γ-proteobacteria (E. coli being the model), whereas the DXP-independent PdxS/PdxT pathway is the predominant route across archaea, fungi, plants, protists, and most other eubacteria (PMID: 17468224, PMID: 19074821). **Animals possess neither de novo pathway** and depend entirely on dietary uptake and salvage (PMID: 17217963).

An alternative route to the same product

The clearest "different molecular means to the same end" is the **PdxS/PdxT complex**, which builds the entire pyridine ring in a single 24-subunit glutamine amidotransferase (12 Pdx1 synthase + up to 12 Pdx2 glutaminase subunits) directly from ribose 5-phosphate, glyceraldehyde 3-phosphate, and glutamine — remarkably producing PLP directly and bypassing PNP and a separate terminal oxidase in the synthetic step (PMID: 17159152, PMID: 20837012, PMID: 17408246, PMID: 17950752). The architectural contrast — six free-standing, mostly recruited enzymes vs. one ornate dedicated machine — is the central axis of variation in de novo B6 biosynthesis.

Evolutionary plasticity: pathways can be rewired

Adaptive laboratory evolution in *Bacillus subtilis* (which natively uses the DXP-independent route) showed that a **truncated, non-native DXP-dependent pathway from E. coli** can be recruited to synthesize PLP: introducing the last two DXP-dependent enzymes plus two genomic alterations restored wild-type-like growth in B6 auxotrophs, exploiting promiscuous "underground metabolism" (PMID: 29027347). Complementarily, E. coli underground metabolism can partly compensate for *pdxB* loss (PMID: 31712440). Together these argue the DXP-dependent route is a **derived, assembleable** solution rather than an ancient, indivisible one.

Cell-biological considerations

In bacteria the pathway is cytoplasmic; there is no well-supported compartmentalization within the module. The main physiological "state" variation is between **de novo synthesis** (precursors abundant, no external B6) and **salvage** (B6 vitamers available), with PdxH as the convergence point. Feedback inhibition of PdxH by PLP provides one layer of homeostatic control (PMID: 34019876).

---

7. Conservation and Origin

Which route is ancestral? The predominance and structural sophistication of PdxS/PdxT, its broad distribution across all three domains of life, and the "assembled-from-recruited-parts" character of the DXP-dependent route together argue that the **DXP-dependent pathway is a derived, lineage-specific elaboration** rather than the ancestral state. Its components are best understood as pre-existing central-metabolism enzymes co-opted into a new pipeline: Epd from the GA3PDH family, PdxB from the D-2-hydroxyacid dehydrogenase superfamily, SerC from serine biosynthesis, and PdxH from salvage. The only innovation unique to the route is PdxJ's TIM-barrel condensation chemistry.

Best representatives of ancestral roles. Where families have expanded, the ancestral function is best studied in the housekeeping paralog rather than the pathway-specialized one: GA3PDH (GapA) for the Epd branch, the serine-biosynthetic PSAT for SerC, and salvage PNPOx for PdxH. The E. coli enzymes remain the reference set for the DXP-dependent route because the biochemistry, genetics, and structures are most complete there. The B. subtilis rewiring experiment (PMID: 29027347) provides direct experimental support for the idea that this route can be assembled evolutionarily from underground metabolism.

---

8. Constraints, Dependencies, and Failure Modes

Ordering constraints (why the sequence is fixed)

1. **Chemical precedence.** Each intermediate is the substrate for the next enzyme; the linear precedence 1→2→3→4→5→6 is fixed by the chemistry.
2. **Substrate phosphorylation gate.** PdxA strictly requires the phosphorylated substrate, so step 4 cannot precede the reactions that install/retain the phosphate (PMID: 12896974).
3. **Two-substrate gate at PdxJ.** PdxJ's catalytic loop closes only when *both* DXP and the aminoketone are bound, preventing unproductive turnover and enforcing convergence of the two arms (PMID: 12269807).
4. **Convergence point.** The E4P arm and the DXP arm are independent until PdxJ; DXP must be supplied by Dxs in parallel.

Self-referential dependency

Because **SerC is itself a PLP-dependent enzyme**, the pathway needs functional PLP to make PLP. This "bootstrap" problem means the pathway cannot be established de novo in a cell with zero PLP without seeding from salvage or a pre-existing pool — an important constraint on the evolutionary establishment of the route and on any synthetic-biology transplant (PMID: 10024454).

Redundancy and bypass

- **Entry redundancy:** Epd and GapA jointly cover step 1; single *epd* mutants retain some PLP, and only *gapA epd* doubles are auxotrophic (PMID: 9696782).
- **Salvage bypass:** Any de novo lesion (e.g., *pdxB*) is rescued by exogenous B6 vitamers via salvage kinases/oxidases (PMID: 27060119, PMID: 32253339).

Diagnostic-gene caveat

Because **PdxH and SerC are shared** (with salvage and serine biosynthesis respectively), and DXP is shared with two other pathways, the presence of these genes does **not** indicate operation of the DXP-dependent de novo route. This is demonstrated in *M. tuberculosis*, which uses the DXP-independent pathway for de novo synthesis yet has bona fide PdxH enzymes operating in salvage (PMID: 26823273). The **diagnostic genes are the early committed ones** — *epd*, *pdxB*, *pdxA*, and *pdxJ*.

Failure modes

- Loss of any committed enzyme (PdxB, PdxA, PdxJ) → B6 auxotrophy unless salvaged.
- In humans, loss of the terminal oxidase orthologue (PNPO) → PNPO-dependent neonatal epileptic encephalopathy, a direct clinical illustration of the terminal step's essentiality (PMID: 32788630, PMID: 34769443).
- Dysregulated PLP homeostasis (e.g., PROSC/PLPBP mutations in humans) underscores the importance of buffering the reactive PLP product, relevant to why bacteria feedback-regulate PdxH (PMID: 27912044).

---

9. Controversies and Open Questions

Strongly supported claims. (i) The pathway comprises six enzymatic steps in the fixed order Epd→PdxB→SerC→PdxA→PdxJ→PdxH, backed by biochemistry, genetics, and multiple crystal structures. (ii) PdxJ is an octameric TIM-barrel that condenses DXP with the aminoketone via a Schiff-base/water-relay mechanism gated by loop closure. (iii) PdxH is a dimeric FMN oxidase performing the terminal O2-dependent oxidation. (iv) The two de novo routes are mutually exclusive and non-homologous, with the DXP-dependent one restricted mainly to γ-proteobacteria (PMID: 11286891, PMID: 15858270, PMID: 17468224).

Points of genuine uncertainty / disagreement. - **Entry-step identity in vivo.** How much of the physiological E4P oxidation is done by Epd vs. GapA, and under what growth conditions, remains only partly resolved; the redundancy complicates clean genetic interpretation (PMID: 7751290, PMID: 9696782). - **Cross-organism extrapolation.** Much of the detailed mechanism is from E. coli, with individual-enzyme structures from *P. aeruginosa* (PdxB) and *M. tuberculosis* (PdxH). Whether kinetic parameters, redundancies, and regulation generalize across all DXP-dependent organisms is not established; mixing data across organisms should be done cautiously. - **Evolutionary polarity.** The DXP-dependent route is inferred to be derived relative to PdxS/PdxT, but the deepest origin is reconstructed, not proven. - **Regulation.** Beyond PdxH feedback inhibition, pathway-level regulation (transcriptional control, metabolite channeling among the E4P-arm enzymes) is not well characterized.

Most important open questions. 1. What controls the relative flux of E4P and DXP into PLP versus their other metabolic fates, and is there channeling among Epd/PdxB/SerC/PdxA? 2. How was the self-referential SerC dependency bootstrapped during the evolutionary establishment of the pathway? 3. Can the reaction intermediates of PdxJ (Schiff base, ring-closure) be trapped to fully resolve the condensation mechanism? 4. How transferable is the E. coli-derived mechanistic model to other γ-proteobacteria and to any non-proteobacterial users?

---

10. Evidence Base

PMID	How it supports / challenges the synthesis
17468224	Establishes the two mutually exclusive de novo routes and the restricted distribution of the DXP-dependent one
21767669	Confirms two de novo pathways + salvage; defines module boundaries
7751290	Biochemically identifies Epd/gapB as the E4P dehydrogenase (step 1)
9696782	Genetic evidence for Epd/GapA redundancy at the entry step
17217963	Places PdxB in the D-2-hydroxyacid dehydrogenase superfamily; names γ-proteobacterial distribution; animals lack de novo synthesis
27060119	Genetic requirement of pdxB; salvage rescue
31712440	Underground metabolism can partly compensate for pdxB loss
10024454	SerC as PLP-dependent fold-type I aminotransferase (Lys198 internal aldimine); dual-function/self-referential
12896974	PdxA structure, Zn^{2+}, strict substrate specificity; defines aminoketone product
9482846	DXP as shared precursor for thiamine, isoprenoid, and pyridoxol; justifies module boundary
11286891	PdxJ homooctamer and ring-closure reaction
12269807	PdxJ multistate binding; DXP-complex structure
12206776	PdxJ two phosphate sites and water-relay mechanism
12686115	PdxJ as key pdx-group enzyme; drug-target rationale
15858270	PdxH terminal reaction, FMN dependence, dual substrate, H_2O_2
16239726	Terminal oxidase conserved bacteria→human
26823273	PdxH operates in salvage in a DXP-independent organism → not diagnostic
17159152, 19074821, 20837012, 17408246, 17950752	Structure/biochemistry of the DXP-independent PdxS/PdxT comparator

PMID	How it supports / challenges the synthesis
29027347	Experimental rewiring of a non-native DXP-dependent route in B. subtilis
32253339, 27912044, 33787481	Salvage and PLP homeostasis context
34019876, 32788630, 34769443	PdxH/PNPO allosteric feedback and human disease

11. Limitations and Knowledge Gaps

- **Organism bias.** The synthesis rests heavily on E. coli, with supporting structures from a handful of other bacteria. Generalization to "all DXP-dependent bacteria" is an inference.
- **In vitro vs. in vivo.** Crystallographic and steady-state kinetic data define enzyme mechanism but not necessarily in vivo flux, channeling, or regulation.
- **No new experimental data.** This review is a literature synthesis; the confirmed findings are drawn from primary and review sources, not from independent bench validation.
- **Sparse quantitative comparison across enzymes.** Only Epd has well-quoted kinetic constants here (Km ≈ 0.96 mM, kcat ≈ 200 s^{-1}); comparable parameters for PdxB, PdxA, PdxJ, and PdxH across organisms were not compiled.

12. Proposed Follow-up Experiments / Actions

1. **Resolve the entry step quantitatively.** Use ^{13}C-flux analysis with defined *epd*, *gapA*, and *epd gapA* mutants to measure each dehydrogenase's fractional contribution to E4P oxidation under different carbon sources.
2. **Trap PdxJ intermediates.** Time-resolved or cryo-trapping crystallography / mass spectrometry of PdxJ with substrate analogs to capture the Schiff base and ring-closure transition state and finalize the condensation mechanism.

3. **Test metabolic channeling in the E4P arm.** Co-purification, crosslinking-MS, and coupled-assay kinetics to determine whether Epd–PdxB–SerC–PdxA form transient complexes that channel unstable intermediates.
4. **Comparative genomics + phylogenetics.** Systematically map *epd/pdxB/pdxA/pdxJ* co-occurrence across bacterial genomes to sharpen the taxonomic boundary of the DXP-dependent route and reconstruct its origin relative to PdxS/PdxT.
5. **Exploit pathway-specific targets.** Since PdxJ (and the committed early enzymes) are absent in humans, screen for inhibitors as antimicrobials against γ-proteobacterial pathogens, using the octameric TIM-barrel active site defined by existing structures.
6. **Probe the self-referential bootstrap.** Engineer PLP-limited strains and reconstitute the pathway from a defined minimal PLP seed to test how the SerC dependency is satisfied during pathway establishment/transplantation.

Prepared as a review-style synthesis of the DXP-dependent de novo PLP biosynthetic module. Claims are anchored to the cited primary literature and authoritative reviews; where evidence is indirect or organism-specific, this is stated explicitly. The dominant caveat is that mechanistic detail derives chiefly from E. coli with a limited set of corroborating structures from other bacteria.
